

IPL Data Analysis — Executive Briefing

17 Seasons of Indian Premier League Data | 2007/08–2024

Project Overview

This project presents a comprehensive **Exploratory Data Analysis (EDA)** of the **Indian Premier League (IPL)**, covering 17 seasons from **2007/08 to 2024**.

The analysis combines **1,095 matches** and **260,920 individual deliveries** to examine the evolution of the tournament, franchise performance, player achievements, match outcomes, venue trends, and strategic factors such as the toss.

Using Python's data-analysis and visualization ecosystem, the project transforms raw IPL data into meaningful insights that highlight the players, teams, venues, and trends that have shaped the league over nearly two decades.

Key Findings at a Glance

Category	Key Finding
 Most Successful Franchise	Mumbai Indians — 144 wins
 Second-Most Successful	Chennai Super Kings — 138 wins
 Leading Run Scorer	Virat Kohli — 8,014 runs
 Leading Valid Wicket Taker	Yuzvendra Chahal — 205 wickets
 Most Player of the Match Awards	AB de Villiers — 25 awards
 Toss Impact	50.83% of toss winners also won the match
 Most Used Venue	Eden Gardens — 77 matches
 Most Matches in a Season	2013 — 76 matches



Methodology

The analysis was performed using **Python** and its leading data-analysis and visualization libraries.



Technology Stack

-  **Python**
-  **Jupyter Notebook**
-  **VS Code**
-  **NumPy** — Numerical analysis and calculations
-  **Pandas** — Data cleaning, transformation, and analysis
-  **Matplotlib** — Data visualization
-  **Seaborn** — Statistical visualization



Datasets Used

The project is based on two primary datasets:

- `matches.csv` — **1,095 match records**
- `deliveries.csv` — **260,920 ball-by-ball records**

This combination enables analysis at both the **match level** and the **delivery level**.



Data Cleaning & Preparation

Before performing the analysis, the datasets were carefully cleaned and prepared to ensure reliable results.

Key preprocessing activities included:

- Removing duplicate records from both datasets.
- Identifying and handling missing values.
- Addressing missing values in important columns such as `method`, `city`, `winner`, and `Player_of_Match`.
- Converting the `date` column into a proper **datetime format** for chronological and season-wise analysis.
- Standardizing relevant team and player information where required.
- Refining wicket calculations to distinguish **bowler-attributed dismissals** from dismissals such as run-outs, retired hurt, and obstructing the field.

- Preparing aggregated datasets for team, player, venue, and season-level analysis.

These preprocessing steps helped improve the consistency and reliability of the subsequent analysis.

Team Performance Analysis

The analysis reveals a highly competitive but historically dominant group of IPL franchises.

All-Time Team Wins

Rank	Team	Wins
1	Mumbai Indians	144
2	Chennai Super Kings	138
3	Kolkata Knight Riders	131
4	Royal Challengers Bangalore	116
5	Rajasthan Royals	112
6	Kings XI Punjab / Punjab Kings	112
7	Sunrisers Hyderabad	88
8	Delhi Daredevils	67
9	Delhi Capitals	48
10	Deccan Chargers	29
11	Gujarat Titans	28

Interpretation

Mumbai Indians emerge as the most successful franchise in terms of total wins, recording **144 victories**, closely followed by **Chennai Super Kings with 138**.

The results demonstrate the long-term consistency of established franchises while also reflecting the changing structure of the IPL and the emergence of newer teams.

Note: Franchise names and rebrands are represented separately in the raw dataset. For example, newer identities such as Royal Challengers Bengaluru and Punjab Kings may appear independently from their historical franchise names.

Standout Player Performances

Top Run Scorers

The batting analysis identifies the following players as the leading run scorers across the dataset:

Rank	Player	Runs
1	V Kohli	8,014
2	S Dhawan	6,769
3	RG Sharma	6,630
4	DA Warner	6,567
5	SK Raina	5,536

Batting Insight

Virat Kohli leads the all-time run-scoring table with **8,014 runs**, highlighting his extraordinary consistency and longevity in the IPL.

Bowling Analysis

Bowling performance was evaluated using delivery-level data, with particular attention given to distinguishing genuine **bowler-attributed wickets** from dismissals that should not be credited to the bowler.

Key Findings

- **R Ashwin** delivered the highest number of balls in the dataset, with **4,661 deliveries**.

- **YS Chahal** leads the valid wicket tally with **205 wickets**.

This distinction provides a more meaningful comparison of bowling impact by excluding dismissals such as **run-outs, retired hurt, and obstructing the field** from bowler wicket statistics.

Player of the Match Analysis

The Player of the Match data highlights several players who have repeatedly delivered match-winning performances.

Rank	Player	Awards
1	AB de Villiers	25
2	CH Gayle	22
3	RG Sharma	19
4	DA Warner	18
5	V Kohli	18

Key Insight

AB de Villiers leads the Player of the Match standings with **25 awards**, demonstrating his exceptional ability to influence match outcomes across multiple IPL seasons.

Does Winning the Toss Increase the Chances of Winning?

One of the most interesting questions explored in this project was:

Does winning the toss give a team a significant advantage?

The analysis indicates that:

- Toss winner also won the match: **50.83%**

- Toss winner lost the match: **49.17%**

Interpretation

The difference is extremely small, suggesting that **winning the toss alone does not guarantee a meaningful competitive advantage**.

Factors such as:

- Team quality
- Player form
- Pitch conditions
- Batting and bowling performance
- Match strategy
- Venue characteristics
- Pressure handling

can have a much greater influence on the final result.

In simple terms, the toss appears to be **closer to a coin flip than a decisive factor**.

Venue Analysis

Venue-level analysis reveals the grounds that have hosted the highest number of IPL matches.

Rank	Venue	Matches
1	Eden Gardens	77
2	Wankhede Stadium	73
3	M Chinnaswamy Stadium	65
4	Feroz Shah Kotla	60
5	Rajiv Gandhi International Stadium, Uppal	49

Venue Insight

Eden Gardens, Kolkata, emerges as the most frequently used venue in the dataset, hosting **77 IPL matches**.

This reflects the importance of major cricket venues in the tournament's long-term history.



IPL Tournament Growth

The number of matches played per season provides an interesting perspective on the evolution of the IPL.

Major Observations

- **2013** recorded the highest number of matches in the analyzed dataset, with **76 matches**.
- **2022 and 2023** each featured **74 matches**.
- **2024** recorded **71 matches**.
- Significant changes in the number of matches correspond with **changes in tournament structure, participating franchises, and competition formats**.



Growth Insight

The IPL has evolved considerably since its inaugural season. Changes in franchise participation and tournament structure have periodically increased the scale of the competition, transforming the IPL from a new cricketing experiment into one of the world's most prominent professional T20 leagues.



Overall Strategic Insights

The analysis reveals several important patterns:

1. 🏆 Long-Term Franchise Dominance

Mumbai Indians and Chennai Super Kings have consistently remained among the strongest franchises based on cumulative victories.

2. 🏏 Individual Excellence

Players such as **Virat Kohli, Rohit Sharma, Shikhar Dhawan, David Warner, and AB de Villiers** demonstrate the importance of consistency and longevity in IPL success.

3. 🎯 Bowling Impact

Yuzvendra Chahal's leading valid-wicket tally demonstrates the impact that specialist bowlers can have over multiple seasons.

4. 🏠 Limited Toss Advantage

The **50.83%** toss-to-match win rate indicates that the toss itself provides only a marginal advantage.

5. 🏟️ Venue Importance

Historic grounds such as **Eden Gardens** and **Wankhede Stadium** have played a significant role in the tournament's development.

6. 📈 Tournament Expansion

Variations in the number of matches across seasons reflect the IPL's evolving franchise structure and competition format.

🎯 Project Conclusion

The analysis of **17 IPL seasons, 1,095 matches, and 260,920 deliveries** provides a comprehensive picture of how the Indian Premier League has evolved over time.

The data highlights the sustained dominance of franchises such as **Mumbai Indians** and **Chennai Super Kings**, the extraordinary batting achievements of **Virat Kohli**, the bowling impact of **Yuzvendra Chahal**, and the match-winning influence of players such as **AB de Villiers**.

At the same time, the analysis challenges some common assumptions. The toss, for example, shows only a **50.83% association with match victories**, indicating that winning the toss alone is not a decisive advantage.

Overall, this project demonstrates how **Python, Pandas, NumPy, Matplotlib, and Seaborn** can transform raw sports data into actionable insights through data cleaning, statistical analysis, aggregation, and visualization.

The IPL is not just a cricket tournament—it is a rich source of data that reveals patterns of performance, strategy, consistency, competition, and evolution.



Project Credits

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Technology: Python • NumPy • Pandas • Matplotlib • Seaborn • Jupyter Notebook • VS Code